

QUALIFICATIONS

- **Software & XR Development:** Unity, Unreal, OpenGL, Blender, XR Development, Real-Time Rendering, Volumetric Video, Multiplayer Networking, Machine Learning for Games
- **Programming Languages:** C#, C++, Python, JavaScript, Java, C, Swift, HTML, CSS, SQL, R, SAS
- **Front-End & UI Development:** Web Development, Three.js, Figma, UI/UX Design, Photoshop, Lightroom
- **AI & Machine Learning:** Reinforcement Learning, AI Agent Training, Python-based AI frameworks (TensorFlow, PyTorch), Prompt Engineering, Real-Time Agent in Games
- **Project Management & Collaboration:** Team Leadership, Git & GitHub, Project Planning, System Design, Problem-solving, Teaching and Mentoring
- **Software & Tools:** Unity, Unreal, Git, Docker, AWS, GCP, R, SAS, Blender, Substance 3D, Figma, Framer, Photoshop, Jira, Trello, Teams, Office, etc.

PROJECTS / EXPERIENCE

Volumetric Video Streaming Framework – IEEE Research Project. [\(DOI\)](#)

Lead Researcher & Developer | 2024

- Developed a cross-platform volumetric 4D video streaming solution using Python and Unity C#, implementing encoding and decoding algorithm, achieving up to 95% reduction in data size and processing over 500 streaming frames with minimal latency.
- Performed thorough performance evaluations, documenting findings and methodologies clearly for academic reporting.
- Published research findings in a peer-reviewed IEEE conference, demonstrating significant contributions to the field of real-time streaming and rendering technologies.

Signed Distance Fields Volumetric Video Compression – IEEE Research Project. [\(DOI\)](#)

Developer | 2025

- Developed a novel compression algorithm for volumetric 4D video using Signed Distance Fields and Singular Value Decomposition, significantly surpassing state-of-the-art methods in compression ratio and visual quality retention.
- Published findings in a peer-reviewed IEEE conference, contributing valuable insights and advancements to the fields of volumetric video compression, real-time rendering, and immersive media technologies.

StudyXR Framework User Research – IEEE Research Project. [\(DOI\)](#)

Lead Researcher & Developer | 2025

- Developed a novel XR-based framework, implementing Meta SDK and AWS, extending traditional VR methodologies, enabling seamless user studies across VR to MR.
- Conducted a user experience study, resulting strong positive ratings.
- Published findings in a peer-reviewed IEEE conference, contributing valuable insights to immersive technology research methodologies and user experience optimization.

Teaching Assistant – Computer Graphics and Game Development

Ontario Tech University, Oshawa, ON | 2024

- Facilitated interactive workshops and lab sessions in Computer Graphics and Game Development, enhancing student engagement and comprehension regarding Unity development, programmable shader language, render pipeline and etc.
- Provided personalized tutoring and mentorship, significantly improving students' understanding of complex technical concepts.
- Collaborated closely with faculty to design rigorous assessments, effectively evaluating student performance and learning outcomes.

StudyVR Framework – IEEE Research Project. [\(DOI\)](#)

Lead Researcher & Developer | 2023

- Led the development of StudyVR, a Unity-based framework to streamline user studies in VR.
- Reduced study setup time by 50%.
- Designed a versatile framework suitable for various research methodologies.

Speed War – GDC and GDW Project. [\(GitHub\)](#)

Lead Programmer & Project Manager | 2022

- Led a team of seven in developing a multiplayer physics-based 3D racing game using Unity C#, featuring TCP/UDP networking protocols for seamless multiplayer gameplay.
- Implemented advanced gameplay mechanics, steering wheel integration, and immersive haptic feedback.
- Designed intuitive UI using Figma and Photoshop, engaging graphical elements, creating a visually appealing and user-friendly gaming environment.
- Managed end-to-end project lifecycle, including planning, task delegation, debugging, and quality assurance, successfully delivering game features.

Flappy Cheese – AI-Enhanced Entertainment Game. [\(itch.io\)](#)

Solo Developer | 2023

- Independently developed an innovative twist on classic flappy-style gameplay using Unity C#.
- Designed and implemented reinforcement learning algorithms to train two adaptive AI agents.
- Managed end-to-end game development lifecycle, including game design, AI integration, user interface (UI), and gameplay optimization.

EDUCATION & CERTIFICATES

Master of Science – Computer Science.

Ontario Tech University, Oshawa ON, 2024 – Present

- Field of study: Human Perception, Avatar Presentation, Immersive Technology, Human-Computer Interaction, Advanced Computer Graphics, AI and Automation, Data Analytics

Bachelor of IT – Game Development & Entrepreneurship.

Ontario Tech University, Oshawa ON, 2019 – 2023

- Field of study: Software Development, Computer Graphics, Computer Mathematics, User Experience Design, Entrepreneurship, Economics, Accounting

Programmer Certificate in Training Certificate of National Information Technology Education Project.

Peking University Beida Jade Bird Group, Shanghai, China, 2014

- Programmer Certificate in Training, National Information Technology Education Project